

Traffic Network Simulator: Final Statistics

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Simulation Overview

The traffic simulation was executed on a modular, custom-built engine utilizing a discrete time-step architecture. The test topology is a planar network consisting of 5 junctions (J1–J5) and 8 directional roads.

- **Traffic Sources:** Junctions J1 and J2
- **Traffic Sinks:** Junctions J3 and J4
- **Queue Management:** Handled on incoming roads
- **Junction Scheduling:** Round-Robin arbitration
- **Routing Protocol:** Static shortest-path routing

Final Metrics

The simulation generated the following throughput and performance statistics based on the vehicles successfully absorbed by the network sinks.

Sink Node	Total Vehicles Absorbed (Throughput)	Average Travel Time (Ticks)
J3	14	7.00
J4	85	5.62

Table 1: Throughput and average travel time metrics recorded at simulation end.